ABSTRACT

AI Cyber Shield – **Intelligent Multi-Modal Phishing and Digital Scam Protection System** is an **Android-based application** designed to help users identify potential **phishing attacks and online scams**. In today's digital world, cybercriminals use fraudulent websites, suspicious links, QR codes, SMS messages, and emails to steal personal information and commit financial fraud. The main objective of this project is to provide a simple and user-friendly system that helps users analyse suspicious content and understand its potential risks.

The system uses **Machine Learning techniques** and **rule-based detection** methods to analyse submitted content and classify potential threats as **SAFE, SUSPICIOUS, or DANGEROUS**. The application supports **URL scanning, QR code scanning, SMS analysis, and email analysis**. It displays the threat status, confidence level, warning indicators, and security advice when available. The application is developed using **Java in Android Studio**, while the **Python Flask backend** handles analysis requests. **Firebase Authentication** is used for user registration and login, and **Cloud Firestore** stores user profile information and scan history. The system also includes a **Cyber Awareness Challenge** to improve users' knowledge of online scams.

The proposed system aims to **increase cybersecurity awareness** and help users make safer decisions before interacting with potentially harmful content. By combining multiple scanning features in a single application, AI Cyber Shield provides a convenient approach to identifying possible digital threats. Although the system cannot guarantee the detection of every scam, it can assist users in recognising suspicious activities. Future improvements may include **larger training datasets, enhanced detection accuracy, real-time threat updates, and support for additional languages.**

Keywords: AI Cyber Shield, Phishing Detection, Digital Scam Detection, Machine Learning, Random Forest Classifier, URL Analysis, QR Code Scanning, SMS Analysis, Email Analysis, Cybersecurity, Firebase, Android Application.